DS4420 Section 1 – Project Literature Review
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From History to Prediction: Estimating Unseen Movie Ratings with Three Machine Learning Approaches

Introduction

The core challenge of recommender systems is predicting how a user would rate content they have never seen, based solely on their past behavior. This project addresses that problem by implementing and comparing three machine learning models — SVD-based matrix factorization, Neural Collaborative Filtering (Neural CF), and Bayesian Probabilistic Matrix Factorization (Bayesian PMF) — on the MovieLens dataset. The literature review below is organized along two dimensions: how different methods tackle the same rating prediction problem, and how the same modeling framework has been applied to related problems, informing our approach from a broader perspective.

Part I: Different Methods, Same Problem

Resnick et al. built GroupLens, one of the first collaborative filtering systems, which predicted ratings using weighted averages of similar users' scores, measured by Pearson correlation [1]. While foundational, this approach breaks down with sparse data and does not scale well, motivating the development of more powerful methods.

Bell and Koren found that explicitly modeling biases in rating data — a global mean, a user bias, and an item bias — produces a simple baseline that outperforms many complex neighborhood methods [2]. This insight directly shapes the bias terms in our SVD implementation.

Koren et al. demonstrated that decomposing the user-item matrix into low-dimensional latent vectors significantly outperforms neighborhood methods [3]. Trained via stochastic gradient descent with L2 regularization, matrix factorization captures hidden preference patterns and handles sparse data naturally. This is the direct theoretical basis for our first model, implemented manually in NumPy.

He et al. argued that the inner product in matrix factorization is inherently linear and cannot capture complex user-item interactions [4]. Their Neural Collaborative Filtering framework replaces the inner product with an MLP, and their NeuMF architecture combines matrix factorization and MLP in parallel. Experiments on MovieLens confirm that non-linear modeling improves prediction accuracy, motivating our second model.

Part II: Same Method, Similar Problem

Salakhutdinov and Mnih extended matrix factorization into a probabilistic framework, treating user and item vectors as random variables with Gaussian priors [5]. Their Bayesian extension, BPMF, applies MCMC for full Bayesian inference over all parameters, achieving better generalization on sparse data and providing uncertainty estimates that point estimates cannot offer. Originally developed for general matrix completion tasks, this framework transfers naturally to movie rating prediction and forms the basis of our required Bayesian model.

Summary

Rating prediction has evolved from similarity-based neighborhood methods, through bias modeling and matrix factorization, to neural and Bayesian approaches. Our three models represent three key points along this trajectory: SVD for classical matrix factorization, Neural CF for non-linear modeling, and Bayesian PMF for probabilistic inference. Comparing all three on the same dataset allows us to directly examine how different modeling assumptions affect predictive performance.

References

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Groupwork GitHub link: https://github.com/TinaChen03/DS4420_project